

GdsFile#

<https://pytorch.org/docs/stable/generated/torch.cuda.gds.GdsFile.html>

Description:

Wrapper around cuFile. cuFile is a file-like interface to the GPUDirect Storage (GDS) API. See the cufile docs for more details. filename (str) – Name of the file to open. flags (int) – Flags to pass to os.open when opening the file. os.O_DIRECT will be added automatically.

Function Signature

```
class torch.cuda.gds.GdsFile(filename, flags)[source]#
```

Parameters

```
deregister_handle()[source]#
```

```
load_storage(storage, offset=0)[source]#
```

Parameters

Code Examples

```
>>> src1 = torch.randn(1024, device="cuda")
>>> src2 = torch.randn(2, 1024, device="cuda")
>>> file = torch.cuda.gds.GdsFile(f, os.O_CREAT | os.O_RDWR)
>>> file.save_storage(src1.untyped_storage(), offset=0)
>>> file.save_storage(src2.untyped_storage(),
offset=src1.nbytes)
>>> dest1 = torch.empty(1024, device="cuda")
>>> dest2 = torch.empty(2, 1024, device="cuda")
>>> file.load_storage(dest1.untyped_storage(), offset=0)
>>> file.load_storage(dest2.untyped_storage(),
offset=src1.nbytes)
>>> torch.equal(src1, dest1)
True
>>> torch.equal(src2, dest2)
True
```

Detailed Documentation

Documentation

Wrapper around cuFile.

cuFile is a file-like interface to the GPUDirect Storage (GDS) API.

See the [cufile docs](#) for more details.

filename (str) – Name of the file to open.

flags (int) – Flags to pass to `os.open` when opening the file. `os.O_DIRECT` will be added automatically.

Deregisters file descriptor from cuFile Driver.

This is a wrapper around `cuFileHandleDeregister`.

Loads data from the file into the storage.

This is a wrapper around `cuFileRead`. `storage.nbytes()` of data will be loaded from the file at offset into the storage.

storage (Storage) – Storage to load data into.

offset (int, optional) – Offset into the file to start loading from. (Default: 0)

Registers file descriptor to cuFile Driver.

This is a wrapper around `cuFileHandleRegister`.

Saves data from the storage into the file.

This is a wrapper around `cuFileWrite`. All bytes of the storage will be written to the file at offset.

storage (Storage) – Storage to save data from.

offset (int, optional) – Offset into the file to start saving to. (Default: 0)